

# A Practical Guide to LLaMA language model

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TOPIC -- LLAMA LANGUAGE MODEL

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Scout: 17 billion active parameter model with 16 experts, context window of 10M, with 109B parameters in total. Maverick: 17 billion active parameter model with 128 experts, context window of 1M, with 400B parameters in total. The Behemoth model was also announced, but was not released. Meta claimed it was a 288 billion active parameter model with 16 experts and around 2T parameters in total; it was still in training when Scout and Maverick were released. Maverick was codistilled from Behemoth, while Scout was trained from scratch. The training data included publicly available data, licensed data, and Meta-proprietary data such as publicly shared posts from Instagram and Facebook and people's interactions with Meta AI. The knowledge cutoff was August 2024. Meta claimed in its release announcement that Llama 4 bested GPT-4o's score on the LMArena AI benchmark. The company also stated that Llama 4's benchmark score was achieved using an unreleased "experimental chat version" of the model that was "optimized for conversationality", which differed from the version of Llama 4 released to the public. LMArena indicated that it would change its policies to prevent this incident from reoccurring, and responded, "Meta's interpretation of our policy did not match what we expect from model providers. Meta should have made it clearer that 'Llama-4-Maverick-03-26-Experimental' was a customized model to optimize for human preference." Some users criticized Meta on social media for its use of a separate model version tailored for benchmarking, and some additionally accused Meta of training Llama 4 on test sets to further boost its benchmark scores, which Meta denied.

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Initial release The first version of Llama (stylized as LLaMA and sometimes referred to as Llama 1) was announced on February 24, 2023, via a blog post and a paper describing the model's training, architecture, and performance. The inference code used to run the model was publicly released under the open-source GPLv3 license. Access to the model's weights was managed by an application process, with access to be granted "on a case-by-case basis to academic researchers; those affiliated with organizations in government, civil society, and academia; and industry research laboratories around the world". Llama was trained on only publicly available information, and was trained at

various model sizes, with the intention to make it more accessible to different hardware. The model was exclusively a foundation model, although the paper contained examples of instruction fine-tuned versions of the model. Meta AI reported the 13B parameter model performance on most NLP benchmarks exceeded that of the much larger GPT-3 (with 175B parameters), and the largest 65B model was competitive with state of the art models such as PaLM and Chinchilla.

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Reception Wired describes the 8B parameter version of Llama 3 as being "surprisingly capable" given its size. The response to Meta's integration of Llama into Facebook was mixed, with some users confused after Meta AI told a parental group that it had a child. The release of Llama models has sparked significant debates on the benefits and misuse risks of open-weight models. Such models can be fine-tuned to remove safeguards, notably by cyber criminals, until they comply with harmful requests. Some experts contend that future models may facilitate causing damage more than defending against it, for example by making it relatively easy to engineer advanced bioweapons without specialized knowledge. Conversely, open-weight models can be useful for a wide variety of purposes, including for safety research. Open Source Initiative head Stefano Maffulli criticized Meta for describing Llama as open-source, saying that it was causing confusion among users and "polluting" the term.

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Applications The Stanford University Institute for Human-Centered Artificial Intelligence (HAI) Center for Research on Foundation Models (CRFM) released Alpaca, a training recipe based on the Llama 7B model that uses the "Self-Instruct" method of instruction tuning to acquire capabilities comparable to the OpenAI GPT-3 series text-davinci-003 model at a modest cost. The model files were officially removed on March 21, 2023, over hosting costs and safety concerns, though the code and paper remain online for reference. Meditron is a family of Llama-based finetuned on a corpus of clinical guidelines, PubMed papers, and articles. It was created by researchers at École Polytechnique Fédérale de Lausanne School of Computer and Communication Sciences, and the Yale School of Medicine. It shows increased performance on medical-related benchmarks such as MedQA and MedMCQA. Zoom used Meta Llama 2 to create an AI Companion that can summarize meetings, provide helpful presentation tips, and assist with message responses. This AI Companion is powered by multiple models, including Meta Llama 2. Reuters reported in 2024 that many Chinese foundation models relied on Llama models for their training.

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Llama 3 On April 18, 2024, Meta released Llama 3 with two sizes: 8B and 70B parameters. The models have been pre-trained on approximately 15 trillion tokens of text gathered from “publicly available sources” with the instruct models fine-tuned on “publicly available instruction datasets, as well as over 10M human-annotated examples”. Meta AI's testing showed in April 2024 that Llama 3 70B was beating Gemini Pro 1.5 and Claude 3 Sonnet on most benchmarks. Meta also announced plans to make Llama 3 multilingual and multimodal, better at coding and reasoning, and to increase its context window. Regarding scaling laws, Llama 3 models empirically showed that when a model is trained on data that is more than the "Chinchilla-optimal" amount, the performance continues to scale log-linearly. For example, the Chinchilla-optimal dataset for Llama 3 8B is 200 billion tokens, but performance continued to scale log-linearly to the 75-times larger dataset of 15 trillion tokens. During an interview with Dwarkesh Patel, Mark Zuckerberg said that the 8B version of Llama 3 was nearly as powerful as the largest Llama 2. Compared to previous models, Zuckerberg stated the team was surprised that the 70B model was still learning even at the end of the 15T tokens training. The decision was made to end training to focus GPU power elsewhere. Llama 3.1 was released on July 23, 2024, with three sizes: 8B, 70B, and 405B parameters.

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Llama 1 models are only available as foundational models with self-supervised learning and without fine-tuning. Llama 2 – Chat models were derived from foundational Llama 2 models. Unlike GPT-4 which increased context length during fine-tuning, Llama 2 and Code Llama - Chat have the same context length of 4K tokens. Supervised fine-tuning used an autoregressive loss function with token loss on user prompts zeroed out. The batch size was 64. For AI alignment, human annotators wrote prompts and then compared two model outputs (a binary protocol), giving confidence levels and separate safety labels with veto power. Two separate reward models were trained from these preferences for safety and helpfulness using reinforcement learning from human feedback (RLHF). A major technical contribution is the departure from the exclusive use of proximal policy optimization (PPO) for RLHF – a new technique based on rejection sampling was used, followed by PPO. Multi-turn consistency in dialogs was targeted for improvement, to make sure that "system messages" (initial instructions, such as "speak in French" and "act like Napoleon") are respected during the dialog. This was accomplished using the new "Ghost attention" technique during training, which concatenates relevant instructions to each new user message but zeros out the loss function for tokens in the prompt (earlier

parts of the dialog).